

Alan Anto

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EDUCATION

- Brandenburg University of Technology** Cottbus, Germany
MSc.in Artificial Intelligence, Grade: 2.2/5 Apr 2025 - Apr 2027
Courses: Cognitive Systems, Image processing, Neural Networks, Explainable Machine Learning
- APJ Abdul Kalam Technological University** Kerala, India
B.Tech(Hons.)in Artificial Intelligence and Data Science, Grade: 9.1/10 Dec 2020 - Jun 2024
Courses: Data Structures, Data Science, Artificial Intelligence, Machine Learning, NLP, Database systems

SKILLS SUMMARY

- Languages:** Python, SQL, C++
- GenAI Framework:** LLM, Transformers, RAG, Prompt Engineering, Qlora, Hugging Face
- Agentic AI Frameworks:** LangChain, LangGraph, BeeAI, crewAI, Google ADK
- ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Numpy
- Databases:** MySQL, VectorDB(ChromaDB & Faiss), GraphDB(Neo4j)
- Cloud & DevOps:** AWS, Azure, Docker, Git, Gradio, REST API, IBM Watsonx
- Soft Skills:** Leadership, Documentation, MS Office, Time Management

EXPERIENCE

- Indian Institute of Information Technology** Kerala, India
AI Engineering Intern Jun 2023 - Aug 2023
 - Fine-tuned pretrained **YOLOv8 (PyTorch)** on a custom-curated dataset of **5,000+** annotated images, optimizing for real-time inference on resource-constrained edge hardware.
 - Achieved **30 FPS** end-to-end inference throughput by exporting models to **ONNX** and applying **PyTorch dynamic quantization**, reducing model size and latency with minimal accuracy loss.
 - Built a custom data augmentation pipeline combining **OpenCV** and **PyTorch transforms**, improving model robustness by **15%** across varied real-world conditions.

PROJECTS

- AutoRAG Intelligence** | Neo4j, LangGraph, ChromaDB, FastAPI, Gradio, Ollama, Docker, Git LINK
 - Engineered a production-grade **hybrid RAG** system for automotive intelligence that reduced hallucinations by **40%** and optimized **routing efficiency** through a 10-node **LangGraph agent pipeline** using **Neo4j** and **ChromaDB**.
 - Built a secure, full-stack Python application achieving **100%** data privacy and **40%** faster response delivery by orchestrating a **Asnyc FastAPI** backend and **Gradio** frontend with local LLM inference.
 - Developed an end-to-end **data ingestion and indexing pipeline**, achieving **faster retrieval** and a **25%** increase in accuracy validated via **RAGAS**, by implementing custom automotive **entity extraction** and a **hybrid indexing** schema in Neo4j.
- Agentic Code-Mate** | Google ADK, Groq, FastAPI, React, pytest, llama, Git, Docker LINK
 - Built a **multi-agent AI code** editor that accelerated developer workflows by **35%** using **Google ADK** and **LLaMA 3.3 70B**, routing user prompts across **7 specialist agents** via a smart Orchestrator.
 - Delivered real-time **token-by-token** streaming by architecting a **FastAPI WebSocket** backend with a **React + TypeScript** frontend featuring syntax-highlighted, copyable code blocks.
 - Containerized the full stack with **Docker Compose**, reducing deployment to a single command across a multi-stage **Nginx-served frontend** and **Python backend**.
- Autonomous Multi-Agent Travel System** | BeeAI, API Tools, Git, Fast API, React, Async LINK
 - Engineered a **multi-agent travel planner**, achieving **100%** automated data synthesis by architecting a hierarchical **BeeAI framework** with specialized expert agents and **handoff protocols**.
 - Reduced user-perceived latency by **65%** for reasoning tasks by developing an **asynchronous FastAPI** backend that streams real-time agent trajectories and tool events via SSE to a **React UI**.
 - Enhanced LLM reliability by implementing strict execution **guardrails** using **BeeAI RequirementAgents** to enforce conditional tool logic and maintain state during complex agent **handoffs**.

PUBLICATIONS

- Autonomous Wheeled Robot for Intelligent Supermarket Restocking** | LINK Jun 2025
IEEE 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST)

CERTIFICATIONS

- PyTorch for deep learning** Professional Certificate | CERTIFICATE Jun 2026
- IBM RAG and Agentic AI** Professional Certificate | CERTIFICATE Dec 2025
- AWS Generative AI for Developers** Professional Certificate | CERTIFICATE Oct 2025
- Microsoft Azure AI Essentials** Professional Certificate | CERTIFICATE Aug 2025
- IBM AI Engineering** Professional Certificate | CERTIFICATE Jul 2025

LANGUAGES

- German:** B1 (Currently pursuing B2)
- English:** C1 (Professional Working Proficiency)